

Object perception and recognition

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So far, we have talked about...

3. Receptor processing

4. Neural processing

Sensation

2. Light transformation

1. Environmental stimulus

Sensation

Colour

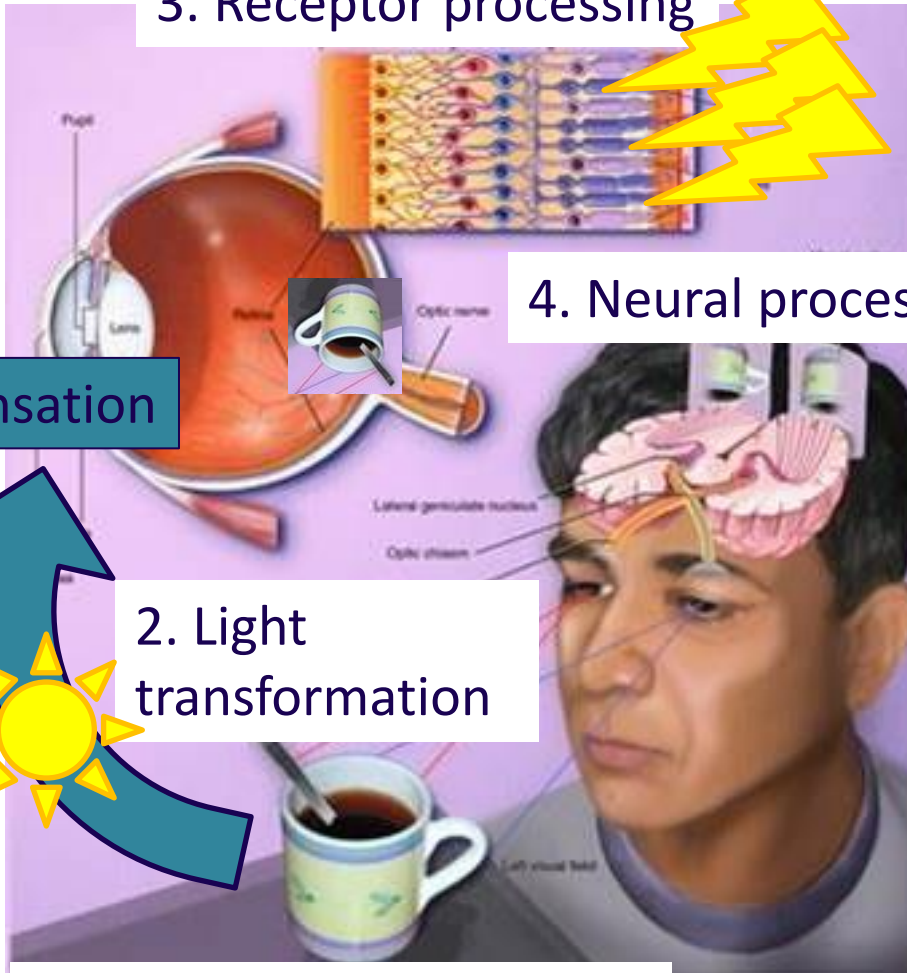
Cones & Opponent neurons

Size & Depth

Retinal image size &
Disparity selective neurons

Motion

Direction-selective & real
motion neurons



So far, we have talked about...

5. Perception

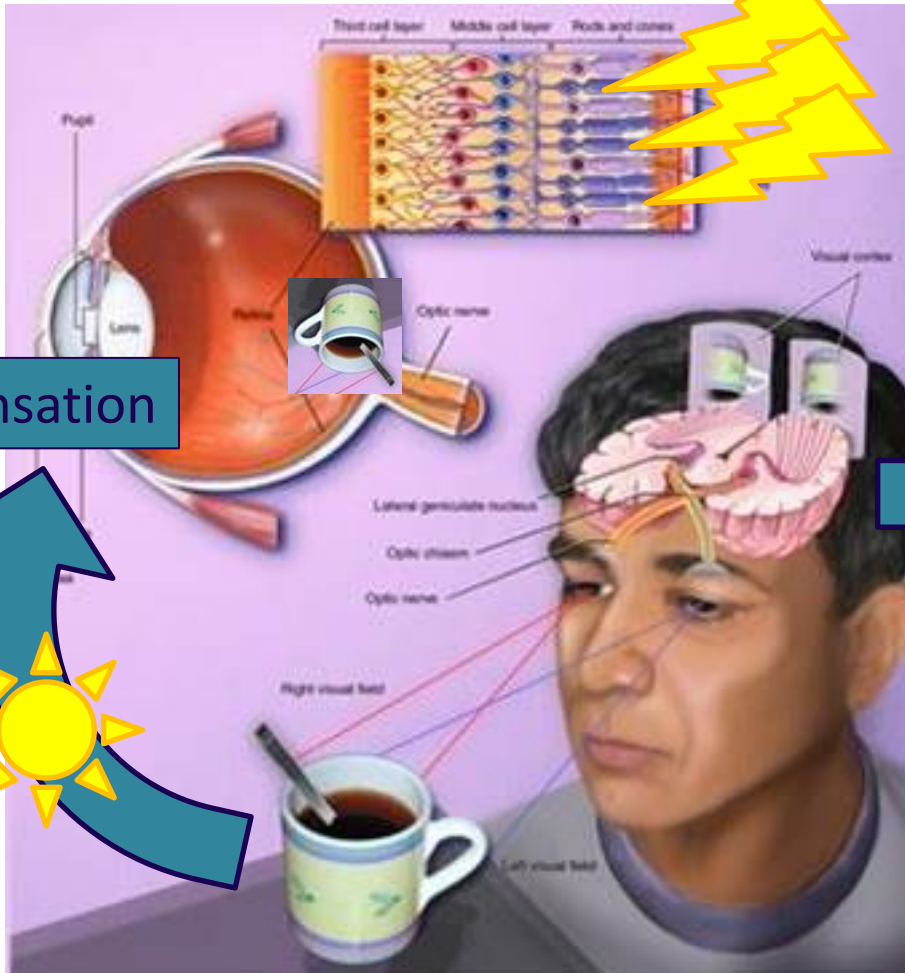


Sensation

Perception

Stimulus guides perception

Bottom-up accounts:



So far, we have talked about...

5. Perception



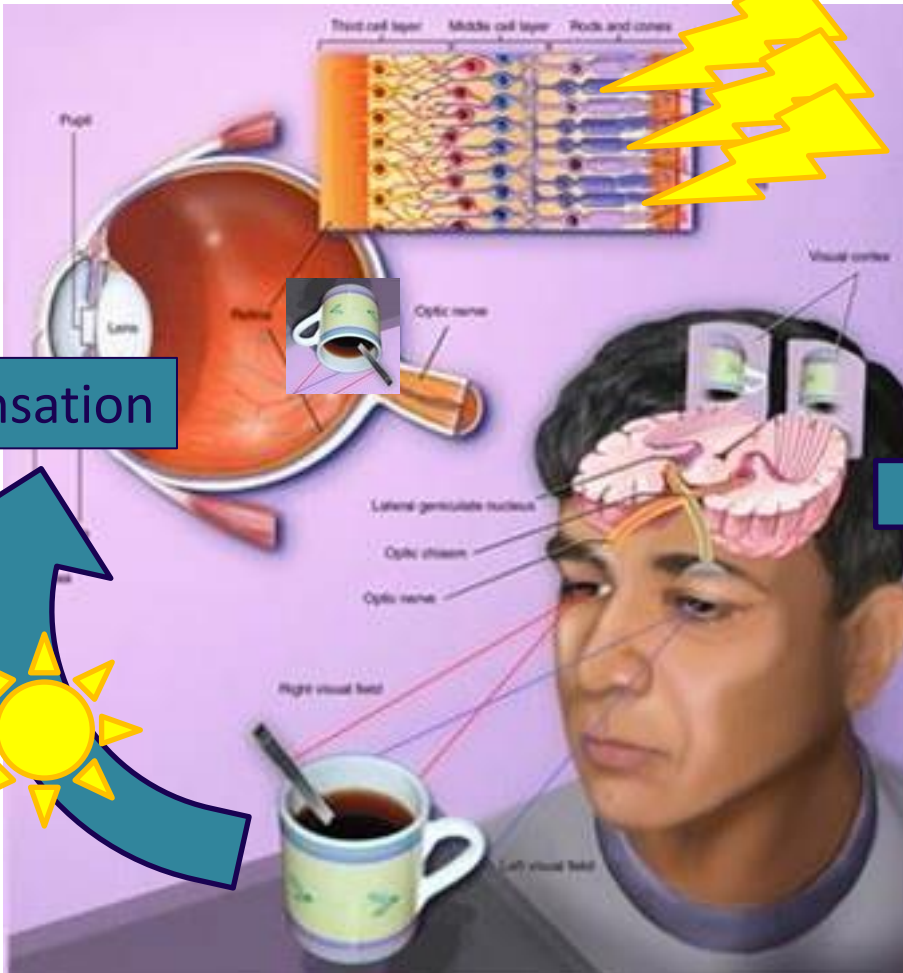
Perception

Stimulus guides perception

Bottom-up accounts:

- Structuralism, psychophysics, & direct perception (Gibson)
 - Colour: Trichromatic and opponent-process theory
 - Motion: Corollary discharge theory
- But not just retinal signal matter...

Sensation



So far, we have talked about...

5. Perception

Cognition
(thinking,
knowledge,
experience)

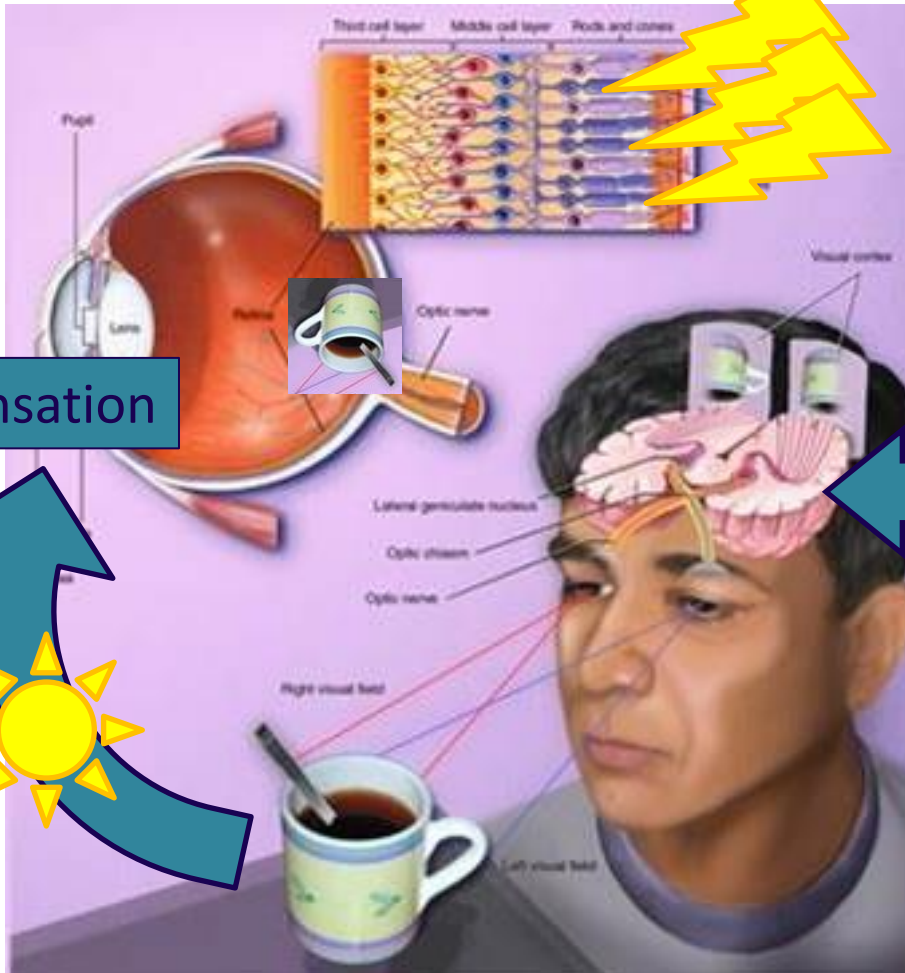


Perception

Cognition guides perception

Top-down accounts:

Sensation



So far, we have talked about...

5. Perception

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(thinking,
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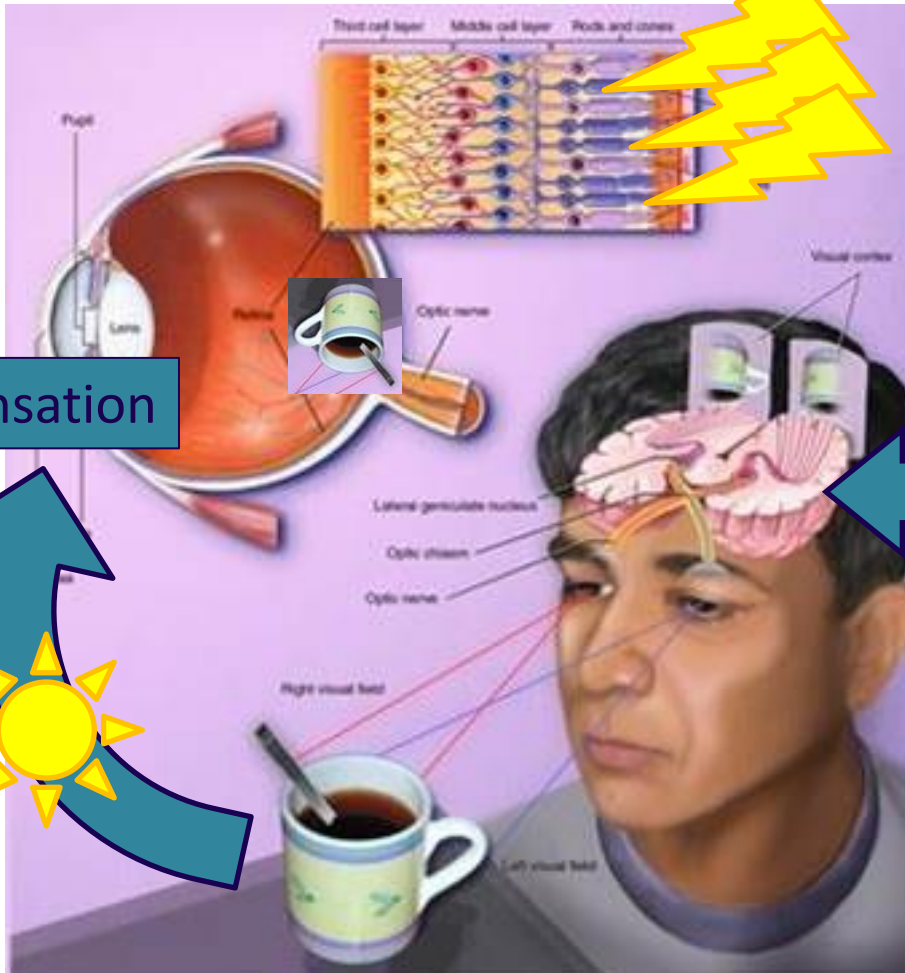
Perception

Cognition guides perception

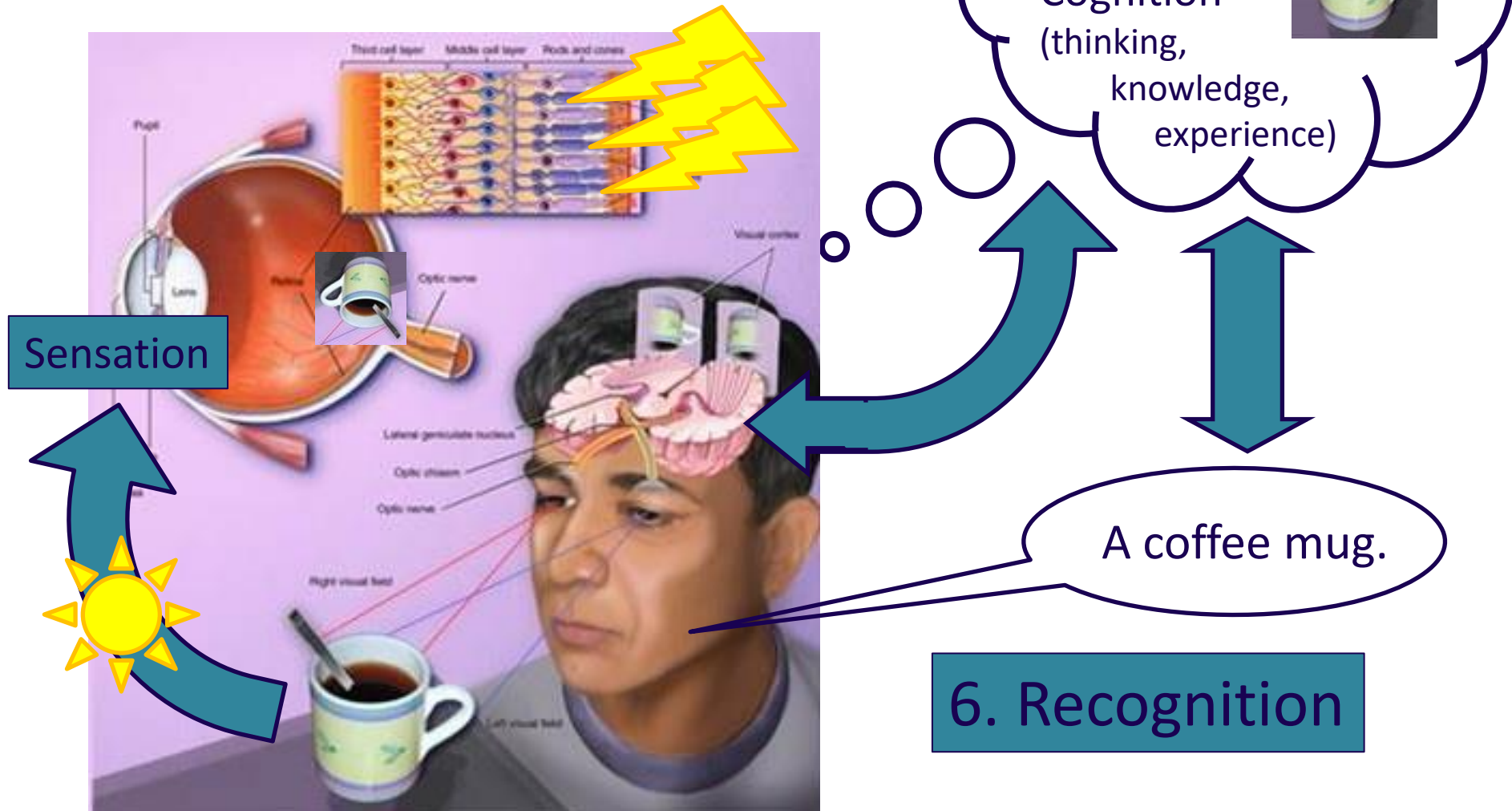
Top-down accounts:

- Gestalt psychology, & constructive perception (Gregory)
- Depth: Pictorial depth cues
- Size: Size-distance scaling equation
- Motion: Biological motion

Sensation



Today...



We previously wanted to know...

- ...how **electrical signals** become a **coffee mug**?



Sensation

The **un-interpreted** sensory impressions created by the detection of stimulus.

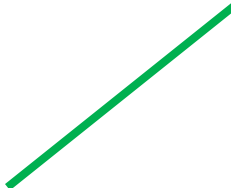


Perception

The (psychological) processes of **making sense** of the sensations.

Today we'll find out...

- ... how we know that it is a **coffee mug**?

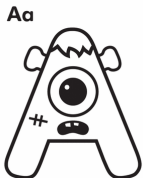
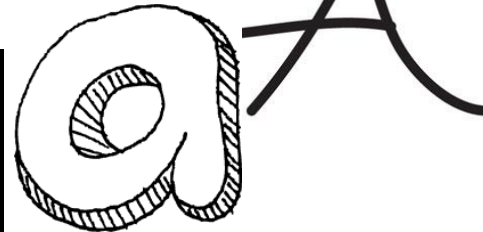
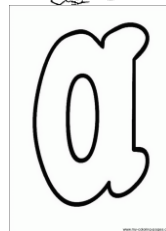
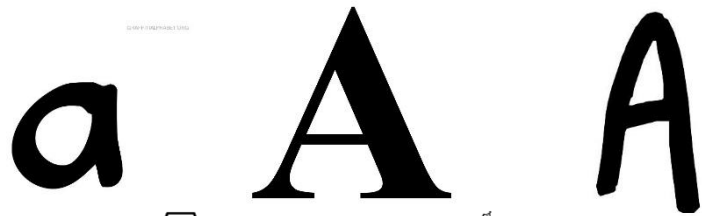
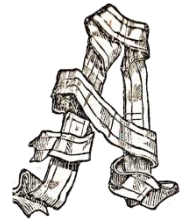


Recognition

The **identification** of a stimulus as an object we are familiar with.

What is the problem?

- An **infinite** number of possible retinal **images** can correspond to a **particular object**.



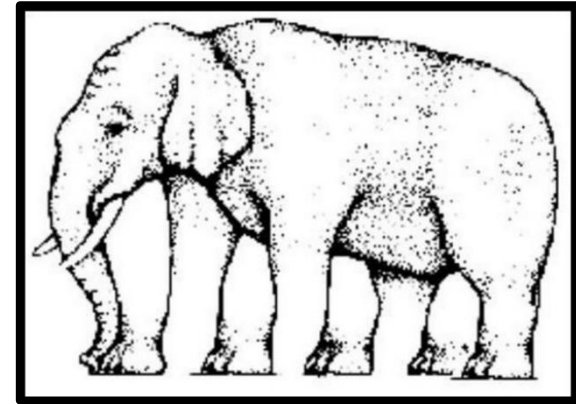
Solution

- Visual input must be matched with **internal object representations** held in object stores.

Imagine an elephant – you can do this, because you activate a visual representation of an elephant in memory.

Unless you have aphantasia....

- Object representation as in
‘the way we know an object looks like’.



If the visual input **matches** the representation
→ object recognition

If the visual input **does not match** the representation
→ no object recognition

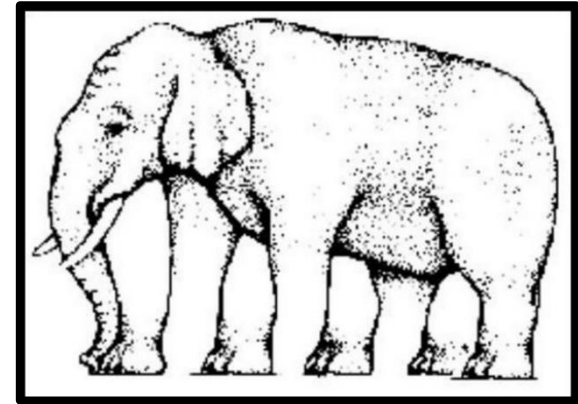
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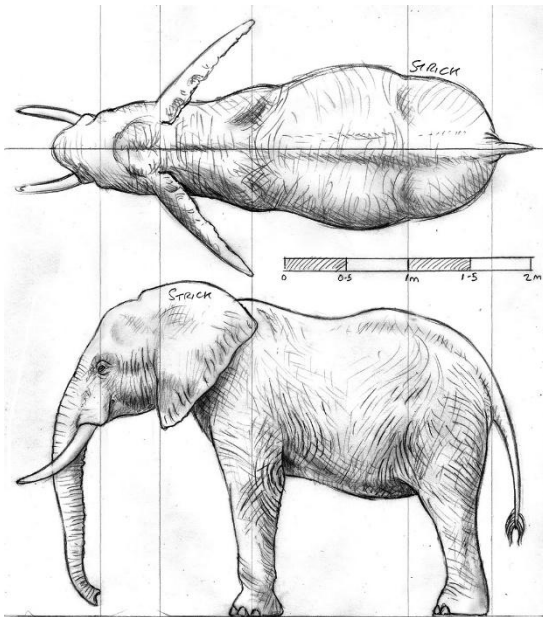
- But how do we do the matching?

Theories of object recognition

- Structural analysis

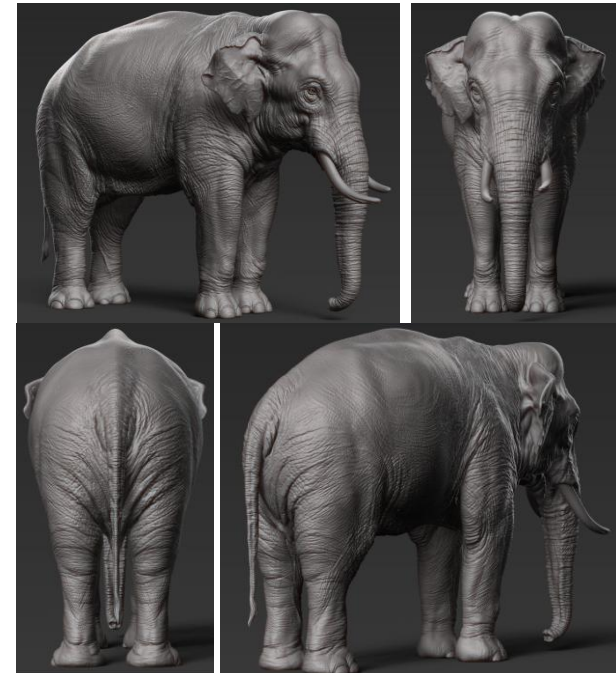
Objects may differ in their 2-dimensional appearance, but they become equivalent in 3D space.

i.e., visual input must be **analysed at different levels**.



2-dimensional

3-dimensional



Structural analysis

- David Marr, 1982



1. Raw primal sketch
2. Full primal sketch
3. 2½-D sketch
4. 3-D model



Structural analysis proceeds through **4 sequential levels...**

...until a **3D model** is achieved...

...which can be **matched with** stored object **representations**.

Marr, D. (2010). Vision: A Computational Investigation into the Human Representation and Processing of Visual Information. Cambridge: MIT Press.

Structural analysis

Initial representation

- David Marr, 1982

1) Raw primal sketch

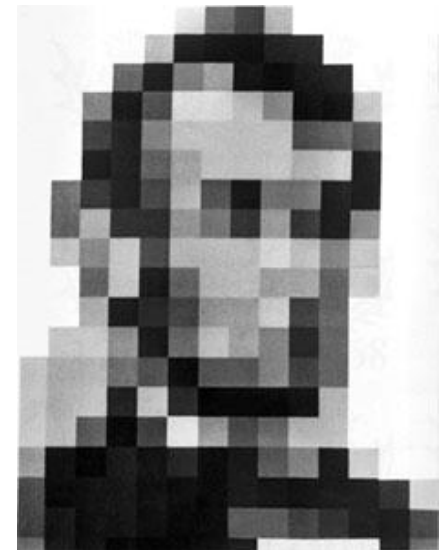
Processing of **intensity** (brightness) **changes** across the retina.



2) Full primal sketch

Geometric organisation of these intensity changes.

i.e., Definition of edges and contours.



Structural analysis

Viewer-centred representation

- David Marr, 1982

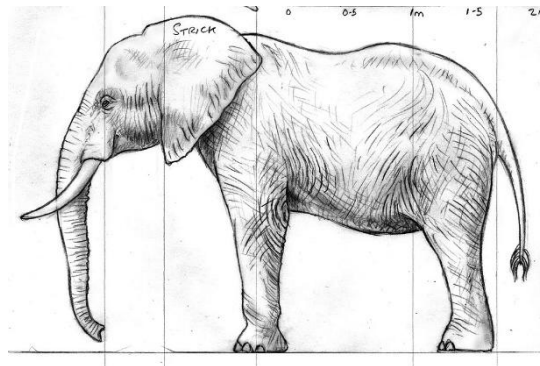
3) 2½-D sketch (½ because it includes depth information)

Processing of **spatial locations** of visible surfaces.

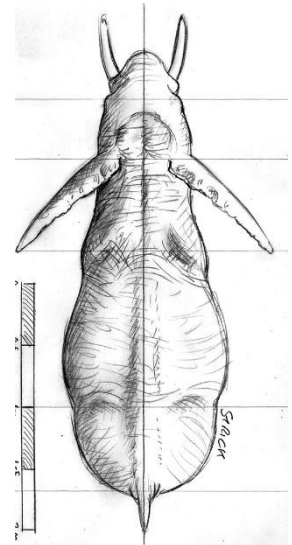
Includes **depth**, but only represents the object **from the observer's viewpoint**.

Perceived objects
cannot be generalised.

But you may still be
able to recognise the
object.



≠



Structural analysis

Object-centred representation

- David Marr, 1982

4) 3-D model (not a sketch anymore)

Viewpoint-independent representation.

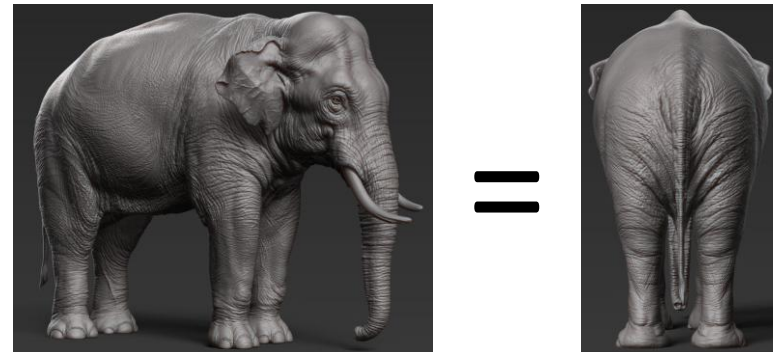
Represents the object independently of the observer's position.

3D standard form is achieved, i.e., real shapes and their relative positions to each other.

Can be matched with stored object representations.

Match → Recognition.

Non-match → Re-analysis.



Structural analysis

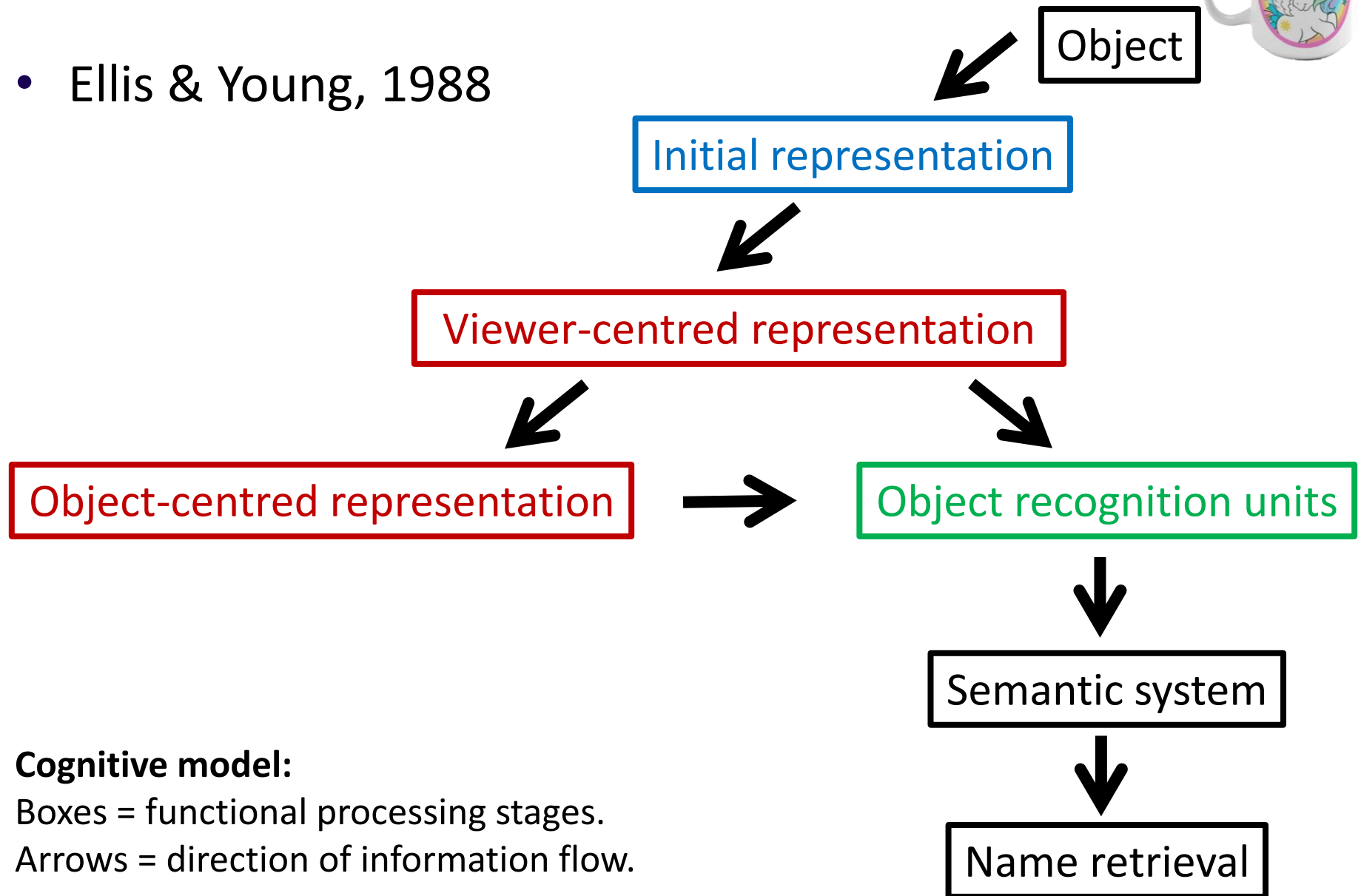
- Why are Marr's hypotheses excellent?
 - They have inspired a **cognitive model**.
- Why is having a cognitive model excellent?
 - It explains a cognitive mechanism which can be used to **make predictions** about human behaviour.
- Why are predictions excellent?
 - We can **test** them and thus **verify or falsify** the model.

The more evidence we find that confirms the model's predictions, the more reliable and valid is the model.

A model of object recognition



- Ellis & Young, 1988



Cognitive model:

Boxes = functional processing stages.

Arrows = direction of information flow.

A model of object recognition



Object



Initial representation

(Primal sketches)

- Ellis & Young, 1988

(Inspired by Marr , 1982)

Analysis of brightness, definition of edges & contours.

A model of object recognition



- Ellis & Young, 1988

(Inspired by Marr , 1982)

Object

Initial representation

(Primal sketches)

Viewer-centred representation

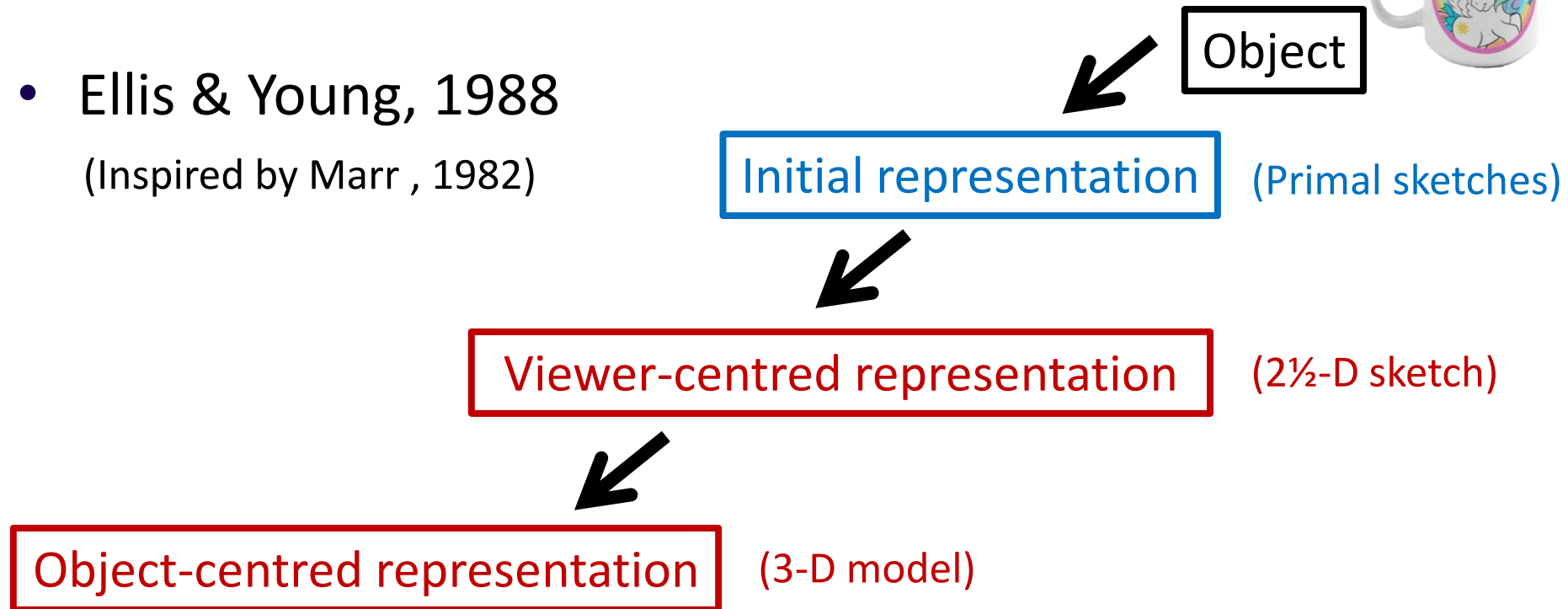
(2½-D sketch)

Edges and contours are assembled to create an object, but only a viewer-centred one, i.e., a representation from a specific angle, from the observer's viewpoint.

A model of object recognition



- Ellis & Young, 1988
(Inspired by Marr , 1982)

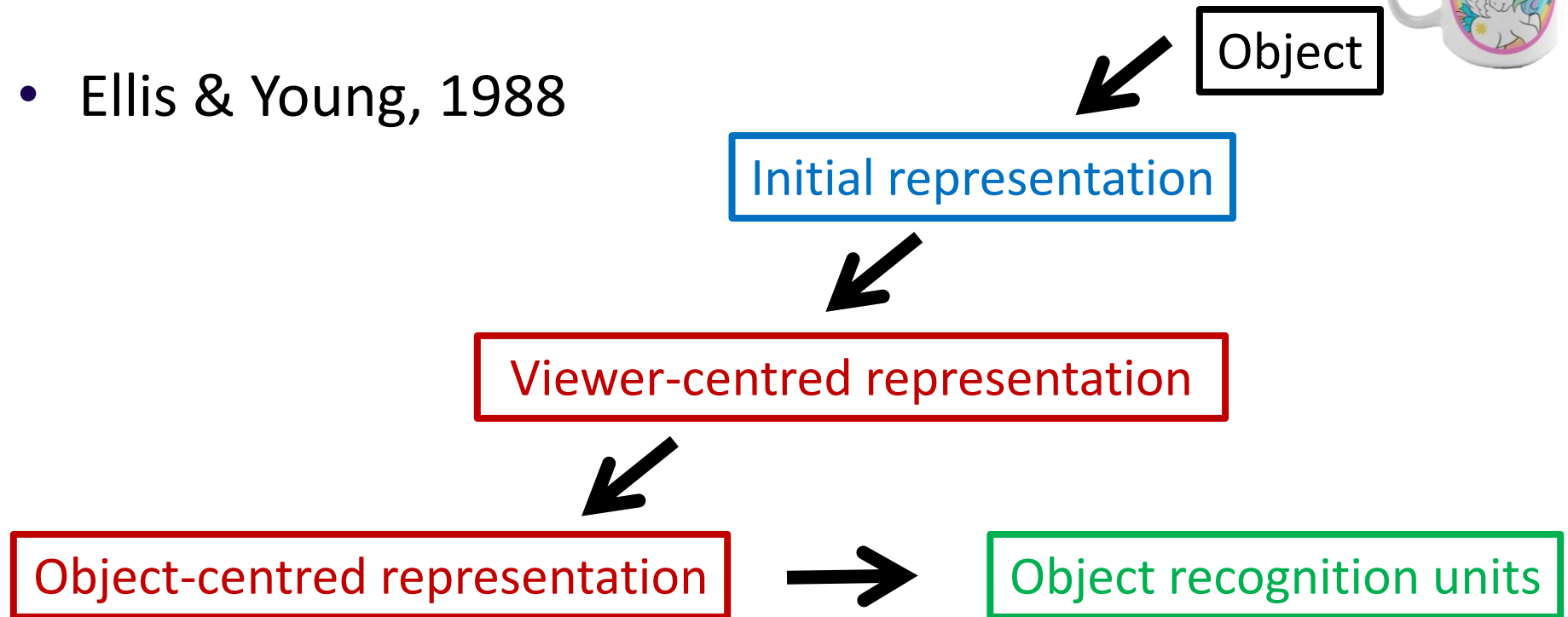


The object's real shape and position, independent of observers viewpoint, i.e., can be matched with stored object representations.

A model of object recognition



- Ellis & Young, 1988

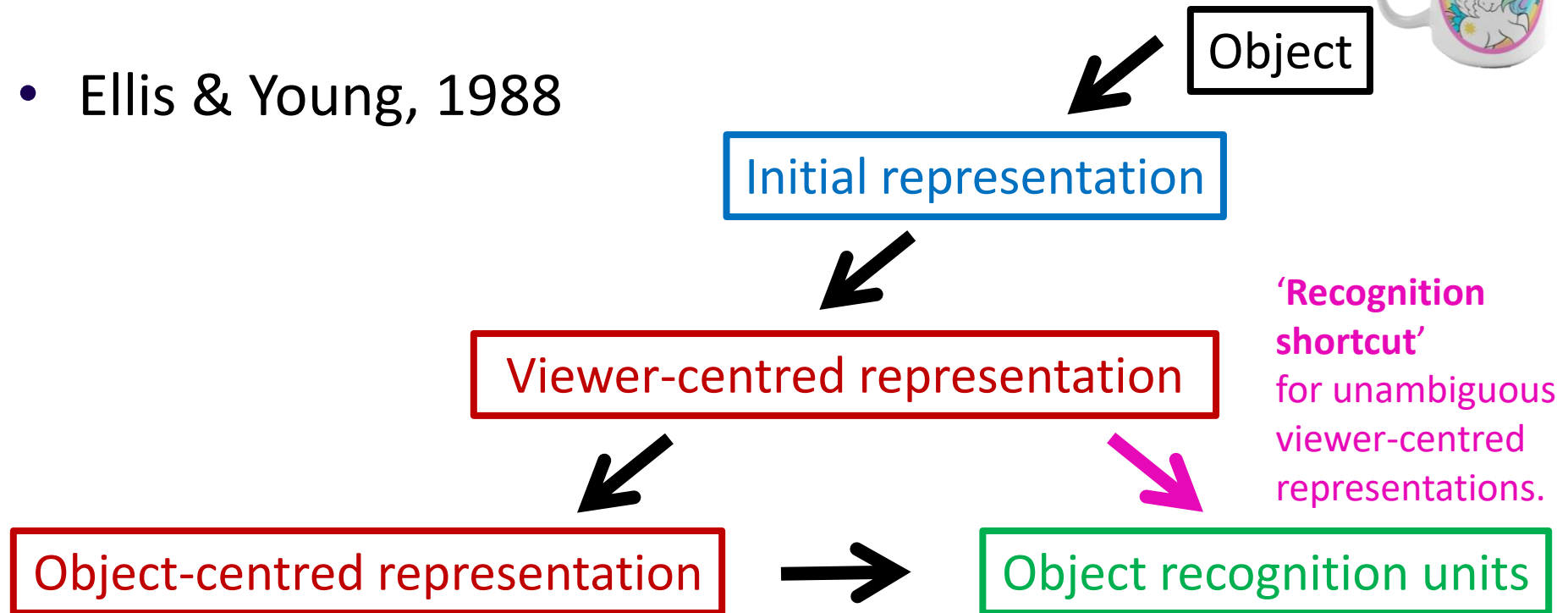


Stored object representations:
Each object has its own unit, i.e., visual object representations stored in memory.

A model of object recognition



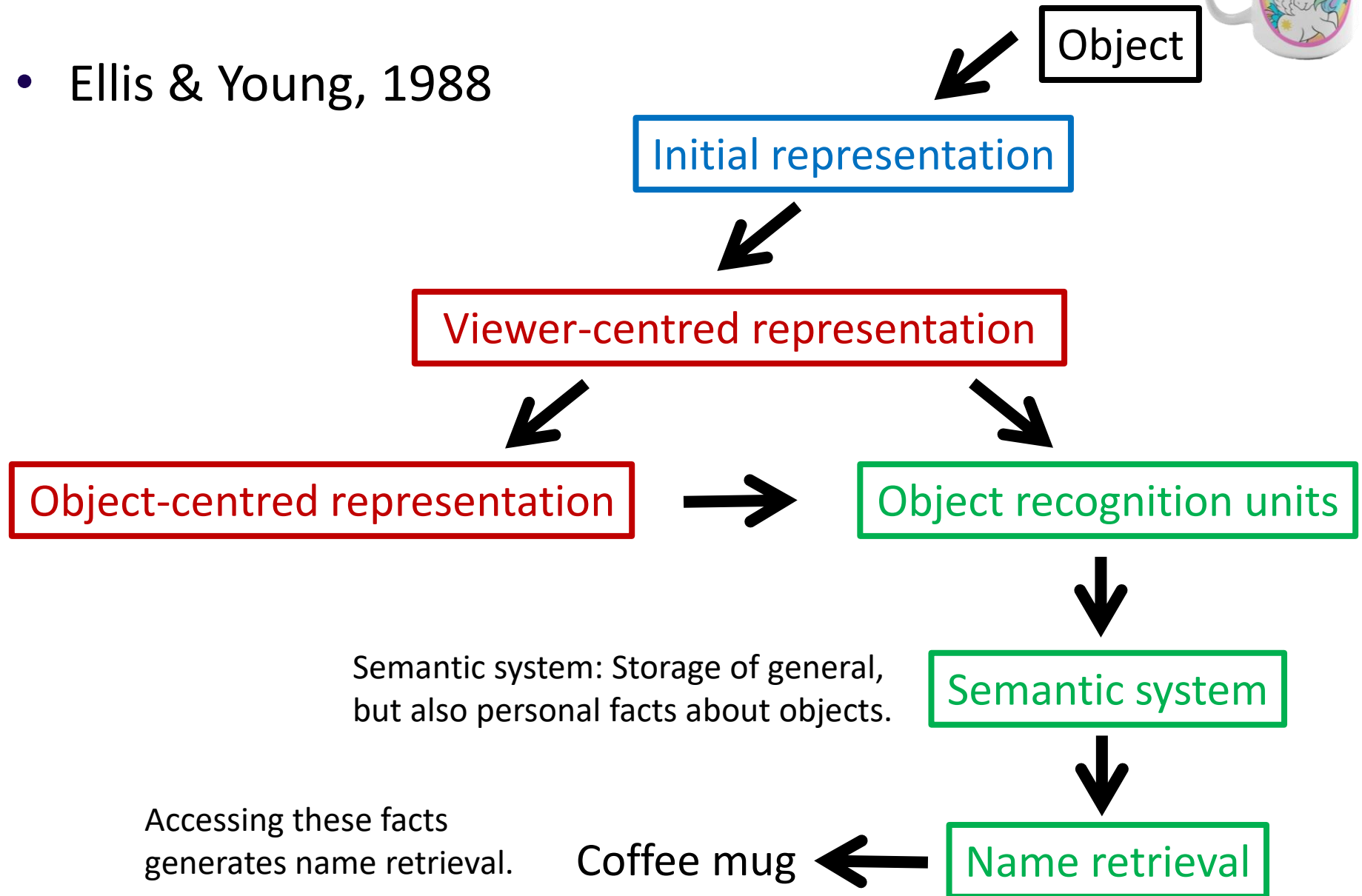
- Ellis & Young, 1988



A model of object recognition



- Ellis & Young, 1988



A model of object recognition



- Ellis & Young, 1988

Sensory processing

Initial representation

Object

perceptual processing

Viewer-centred representation

Object-centred representation

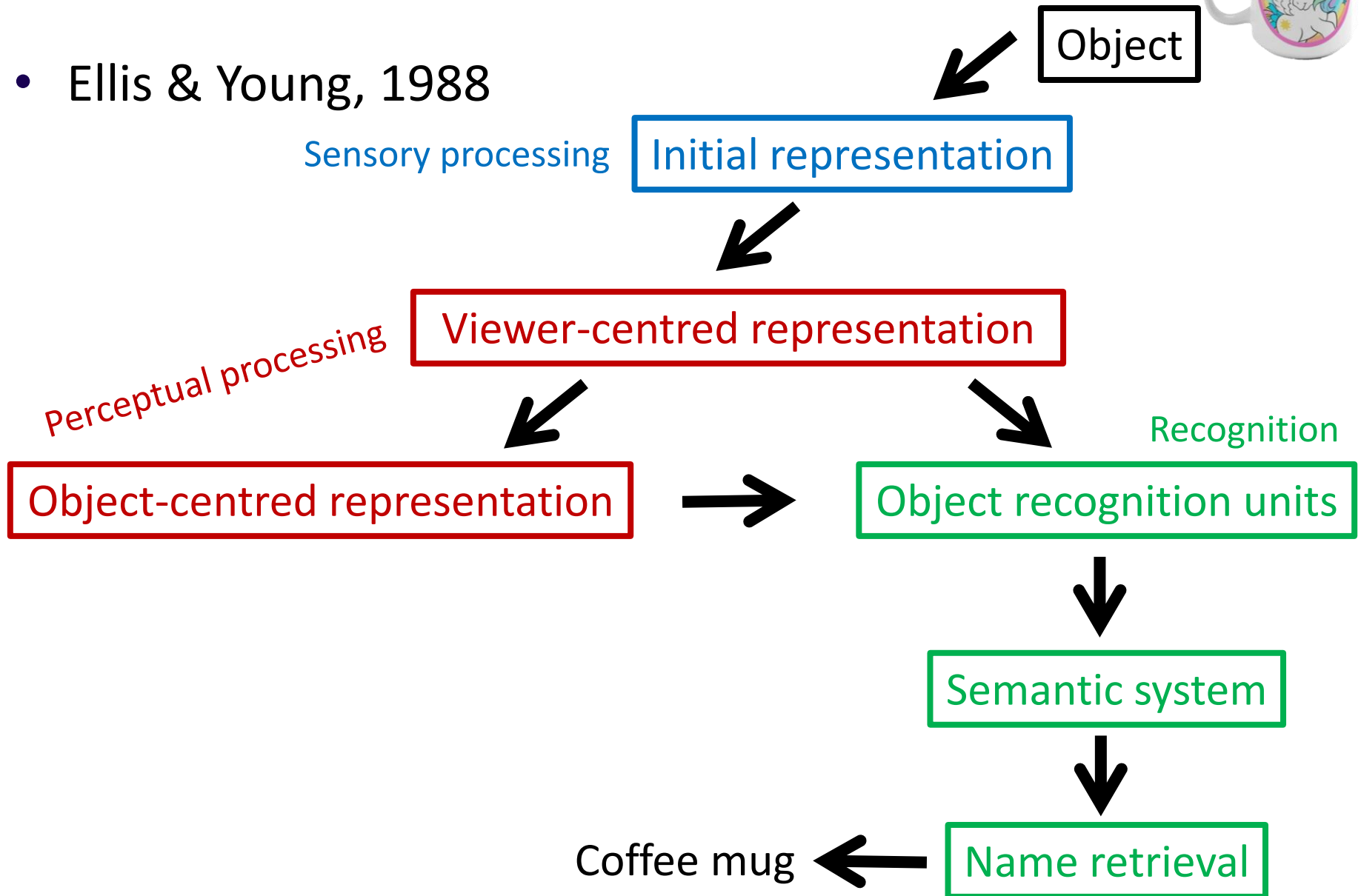
Recognition

Object recognition units

Semantic system

Coffee mug

Name retrieval



A model of object recognition

- Why are Marr's hypotheses excellent?
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- Why is having **a cognitive model** excellent?
 - It explains a cognitive mechanism which can be **used to make predictions** about human behaviour.
- Why are predictions excellent?
 - **We can test** them and thus verify or falsify the model.
The more evidence we find that confirms the model's predictions, the more reliable and valid is the model.

Testing model predictions

- **Mr S**

Reported by Benson & Greenberg, 1969

- Young soldier who suffered carbon monoxide poisoning.
- Neuropsychological tests reveal that he **can**
 - maintain fixation, name colours, and describe sensations.
e.g., a safety pin is “silver and shiny like a watch or a nail clipper”.
 - detect small changes in brightness and wavelength on psychophysical tests.
 - detect movements of small objects.

Testing model predictions

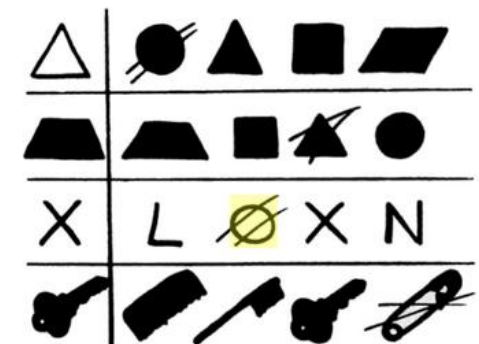
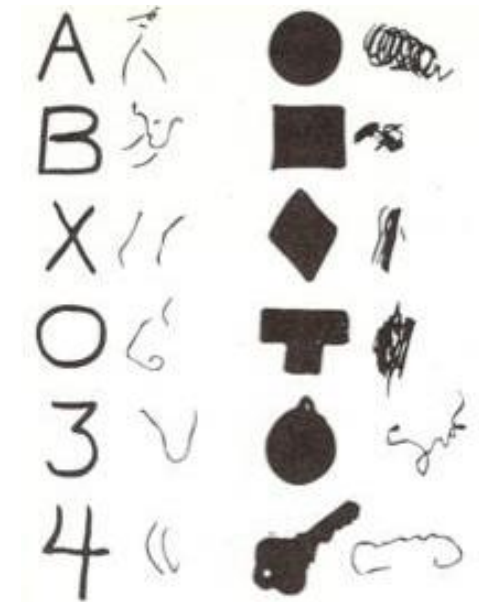
- **Mr S**

Reported by Benson & Greenberg, 1969

- But he **cannot**
perceive shape or form.

e.g., cannot copy simple figures,
letters, digits.

e.g., cannot match simple objects.



Testing model predictions

- **Mr S**

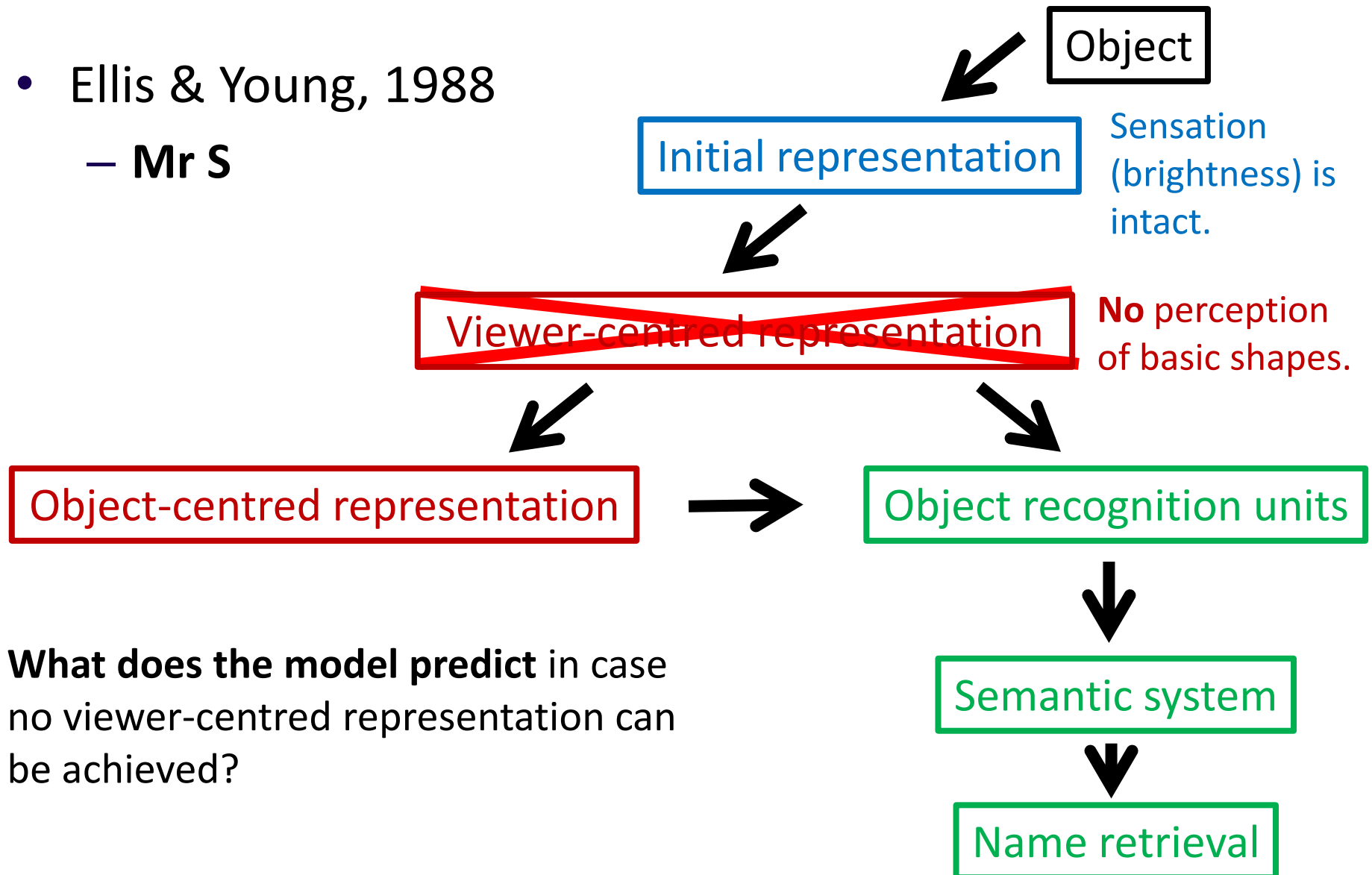
Reported by Benson & Greenberg, 1969

- He also **cannot**

recognise or name objects and pictures of objects (body parts, letters, numbers, faces).

Testing model predictions

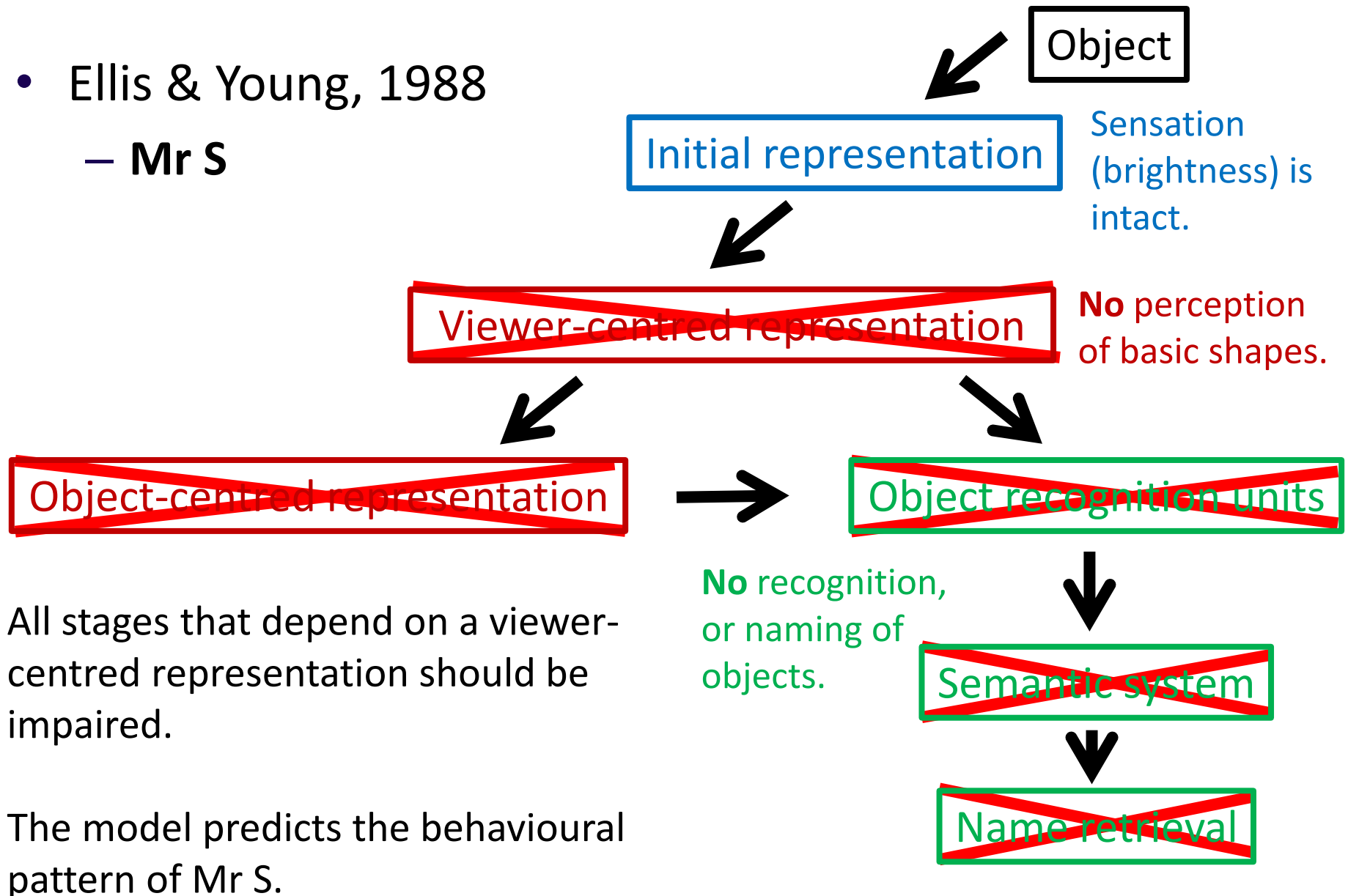
- Ellis & Young, 1988
 - Mr S



What does the model predict in case no viewer-centred representation can be achieved?

Testing model predictions

- Ellis & Young, 1988
 - Mr S

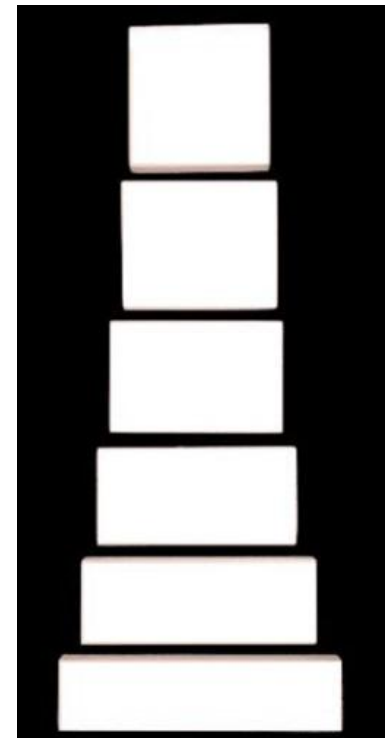


Testing model predictions

- Group of patients with **right posterior lesions**

Reported by Warrington, 1982

- **Can** match objects for size, colour, brightness.
e.g., Efron (1968) rectangle matching.



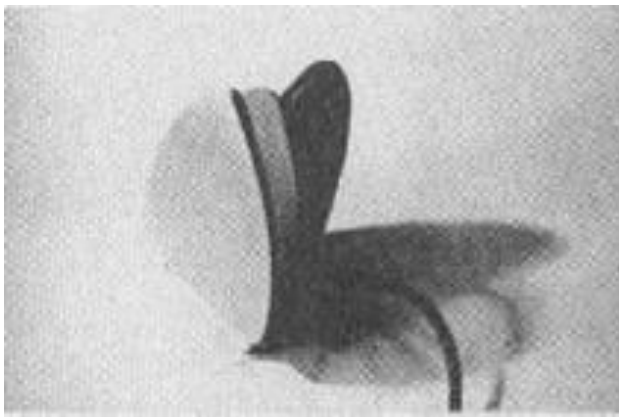
Which square matches the test square?

Testing model predictions

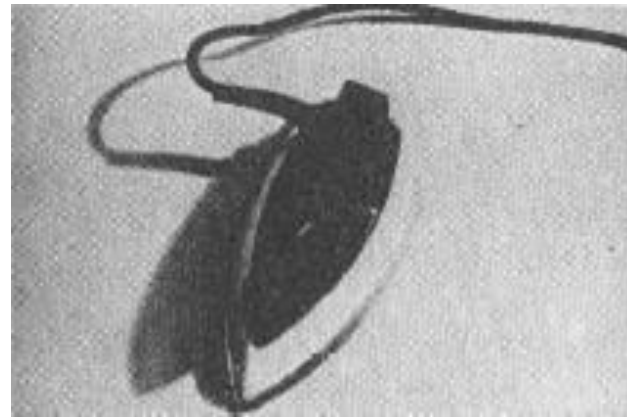
- Group of patients with **right posterior lesions**

Reported by Warrington, 1982

- **Can** recognise objects from pictures in usual views.
- **Cannot** recognise objects in unusual views.



Iron – unusual view



Iron – usual view

Testing model predictions

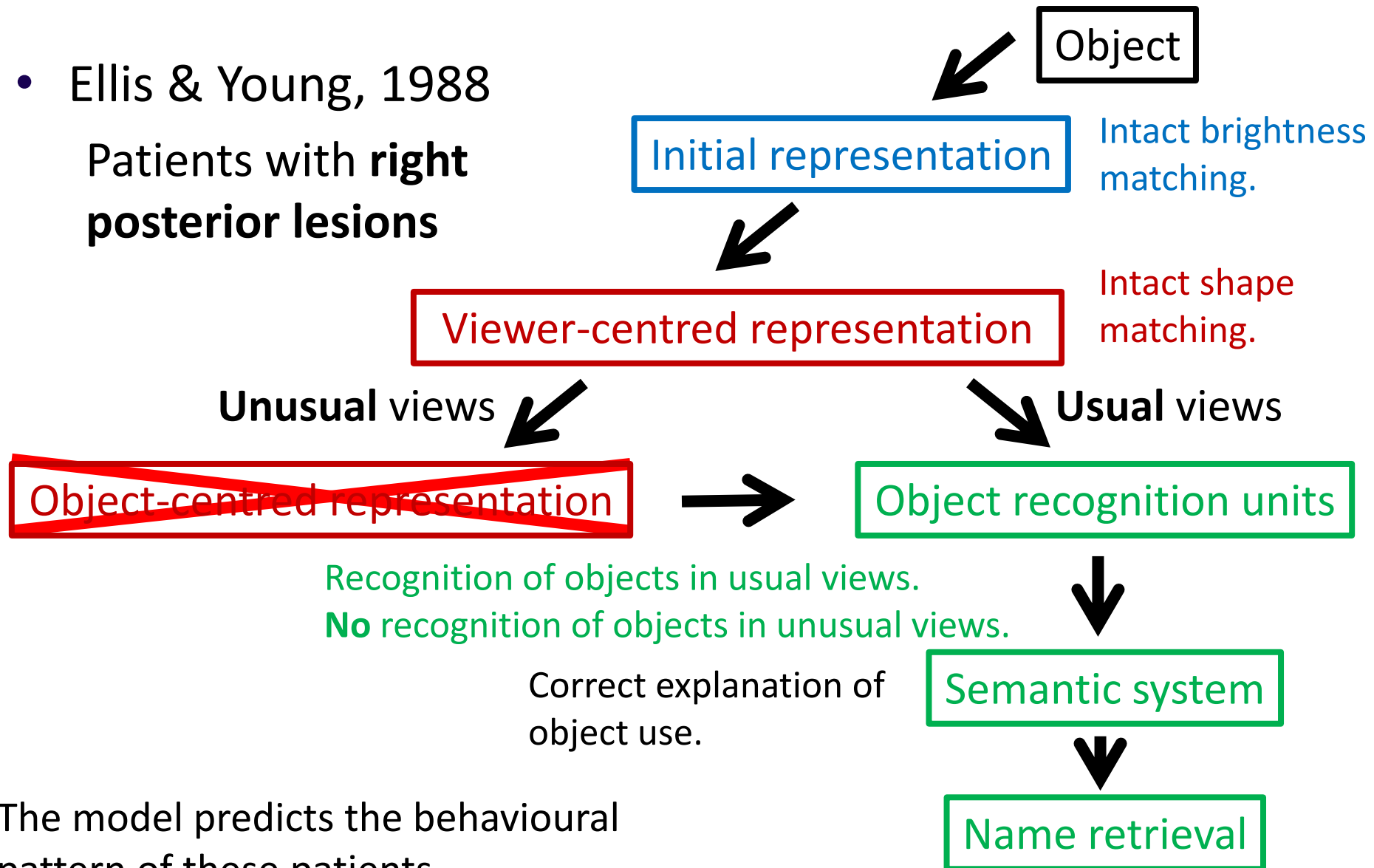
- Group of patients with **right posterior lesions**

Reported by Warrington, 1982

- **No impairments** with object recognition in everyday life.
- **Can** explain use of objects.
 - e.g.,
 - “Torch = hand-held light”.
 - “Compass = tools for telling direction you are going”.

Testing model predictions

- Ellis & Young, 1988
Patients with **right posterior lesions**



The model predicts the behavioural pattern of these patients.

Testing model predictions

- **AB**

Reported by Warrington, 1975

- Civil sergeant with progressive cerebral atrophy (ongoing loss of neurons and connections between them).
- The patient **can**
 - distinguish basic shapes.
 - identify colours, numbers and letters.
 - match pictures of different views (usual and unusual).



Testing model predictions

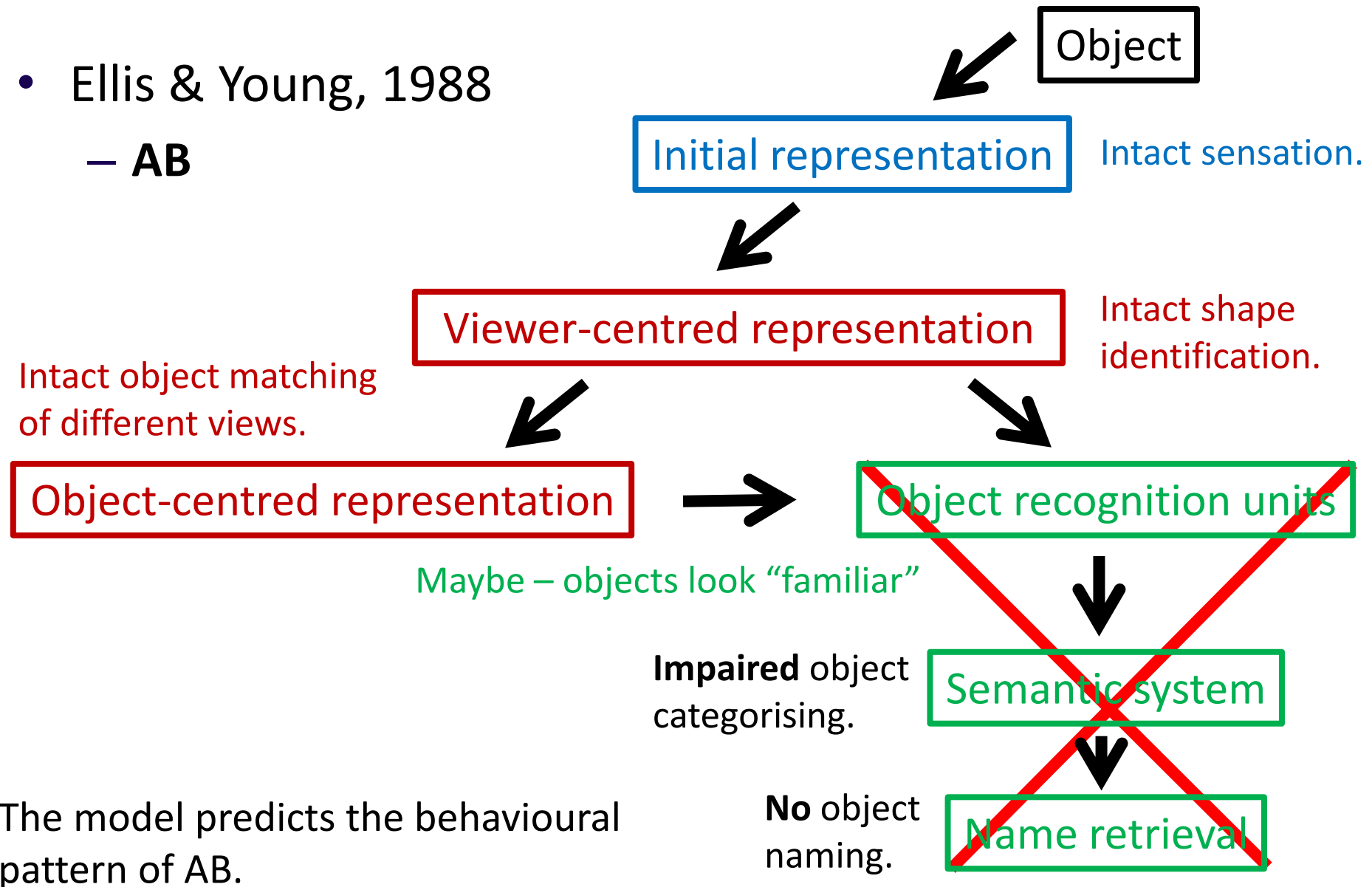
- **AB**

Reported by Warrington (1975)

- **Cannot** name common objects or famous faces.
e.g., photographs of the faces of contemporary personalities.
But states that they look “familiar” but he had “forgotten” them.
- **Makes** substitutions when categorising objects.
e.g., a donkey is a horse, a dog is a cat.
- **Cannot** identify meaningful sounds.
e.g., telephone ringing, dog barking.

Testing model predictions

- Ellis & Young, 1988
 - **AB**



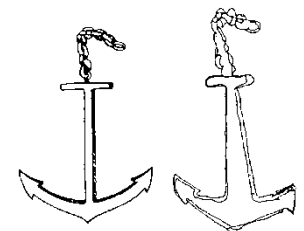
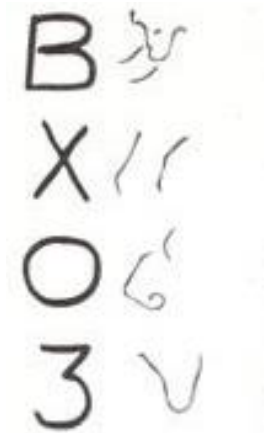
Ellis & Young (1988) model of object recognition

- Evaluation
 - The Ellis & Young model (1988) predicts the behavioural pattern of patients with **two different types of visual agnosia**.
 - Visual agnosia = inability to recognise seen objects.
 - Impaired **perceptual** stages (**apperceptive** agnosia)
 - Impaired **semantic** stages (**associative** agnosia)

Distinction originally suggested by Lissauer, 1890.

Ellis & Young (1988) model of object recognition

- Evaluation
 - **Apperceptive agnosia** (impaired perception)
e.g., Mr S
Cannot copy or match even simple shapes.
Profound **impairment in shape perception**.
 - **Associative agnosia** (impaired semantics)
e.g., AB
Can copy and match objects.
But intact **percept has no meaning**.



ORIGINAL

COPY

Ellis & Young (1988) model of object recognition

- Evaluation
 - How about the patients with **right posterior lesions**?
They cannot create object-centred representations.
They have a **perceptual categorisation deficit**, but don't actually have visual agnosia (object recognition is intact).

Ellis & Young (1988) model of object recognition

- Conclusion
 - The model is **valid**.
i.e., it makes **accurate predictions** about behavioural impairments.
 - But...
...are single cases valid measures to test models?

Ellis & Young (1988) model of object recognition

- Conclusion
 - A cognitive model needs to explain
 - The **functional components** of a mechanism and how they are linked.
 - The **patterns of impairments** observed.
 - The patterns of impairments **not yet observed** (predictions).
 - Observation of only one case for which the predictions do not hold is required for revision (or abandonment).

The Ellis & Young model is valid – **FOR NOW!**

Summary

- We have looked at
 - Three theories of object recognition which make different suggestions about **how visual input is matched with stored object representations**.
 - Template matching** (match against complete object templates),
 - feature analysis** (match against individual object features),
 - structural analysis** (match on different levels of processing, e.g., Marr).
 - Marr's theory of object recognition.
 - Structural analysis** proceeds through **4 sequential levels** (raw & full primal sketch, 2½-D sketch, 3-D model) until a **standard object model** is achieved which **can be matched** with stored object representations.

Summary

- We have looked at
 - Ellis & Young's model of object recognition.

After sensory (**initial representation**) and perceptual (**viewer-centred, object-centred representations**) processing, object models are compared to stored object representations (**object recognition unit**). If the match is successful, semantic information and object names are retrieved.

A **valid model** that accurately predicts different types of object recognition impairments (**apperceptive & associative agnosia, object categorisation deficits**).

Test yourself – learning goals

- After reviewing this lecture you should be able to
 - Explain template matching, feature analysis, and structural analysis approaches to object recognition.
 - Explain David Marr's theory of object recognition.
 - Discuss the different stages involved in object recognition according to Ellis and Young and for each of them explain the consequences of impairments.
 - Define agnosia, name two forms of agnosia and explain how they differ.

Test yourself – multiple choice

1. Which of the following statements about object recognition is wrong?

- a) The basic idea of object recognition is that a perceived object is matched against stored object representations held in memory.
- b) If a perceived object can be named it is recognised.
- c) If a perceived object matches an object recognition unit it is recognised.
- d) In order to recognise an object, sensory processing stages must be fully and perceptual processing stages must at least be partially completed.

Test yourself – multiple choice

2. People with associative agnosia can:

- a) Perceive objects.
- b) Recognise objects.
- c) Name Objects.
- d) Explain the use of objects.

Test yourself – multiple choice

3. People who cannot achieve object-centred representations (according to Ellis & Young, 1988) cannot:

- a) Perceive objects in general.
- b) Match objects according to their brightness or colour.
- c) Recognise objects in general.
- d) Match objects shown from usual and unusual angles.

See you on Monday – Revision Lecture

